2019-2020
Second Semester

University of Bahrain
College of Engineering
Department of Electrical and Electronics Engineering

Smart streetlights

Motivation and Objective

Motivation: In Bahrain, streetlights are being controlled using time and light sensors, which are not very accurate because they do not detect changes in the environment or motion. This method is inefficient because the lights only turn on and off without any awareness of the surroundings. For instance, if there is no traffic or pedestrians, there is no need for the lights to be on.

The objective: To design and implement an advanced streetlights system that utilizes real-time sensor data to accurately detect traffic patterns on the road, and intelligently adjust the light intensity levels accordingly. The aim is to create a highly efficient and effective system that maximizes energy savings while maintaining optimal visibility and safety for drivers and pedestrians.

Design and Implementation

Actual road design: we used a foam board as the base for our road design instead of plywood, which is a heavier material. The foam board was painted with a colorful design depicting various buildings and streets to create a more realistic and vibrant environment.

Motion sensors & zoning: The design of the project requires using eight motion-sensors in total, four for each side of the highway, to detect any movement in the motion sensors' range.

Control unit: It is consisting of two modes of operation:

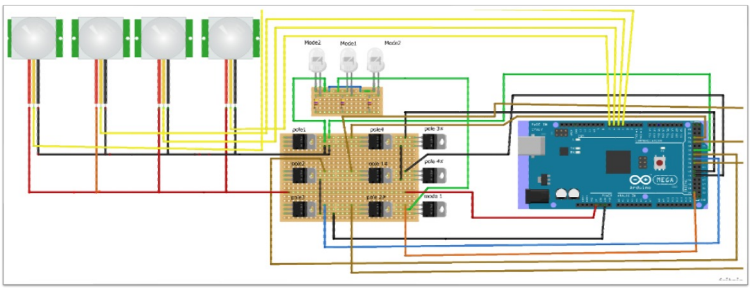
- **Always on stage** providing continuous lighting for drivers to see what's ahead of them.
- **Second stage** is the most significant one, which activates when a car is within range of the motion sensor.

This helps conserve energy and reduce costs by only illuminating the necessary areas at the appropriate time.

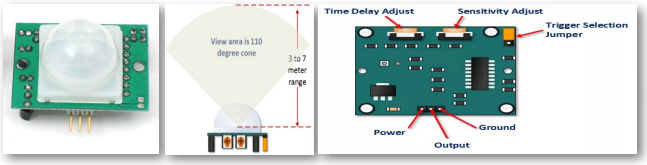
LDR: Light Dependent Resistor is a component in our circuit that detects and identifies the presence of light.

Arduino: All electronic and electrical devices was connected to it which serves as the brain of each device. The choice of controller depends on the specific operations required by the project, as well as the number of inputs and outputs needed. Arduino was an optimal choice for this project due to its suitable memory size and ability to handle the necessary inputs and outputs.





Complete Circuit diagram

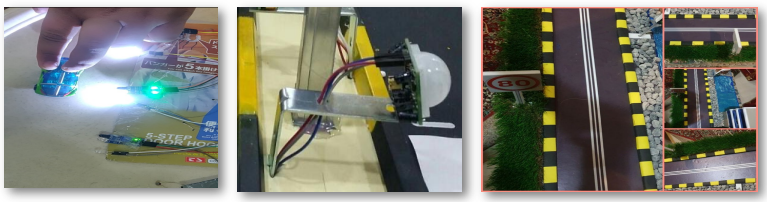


Motion sensor | HR-SR501 Pinouts and controls | LDR

Results & discussion

We have developed a system that is connected to a server and can be controlled automatically or manually using buttons to turn individual pole lighting "ON" or "OFF". Each pole can also be controlled separately, which is useful in case of any problems or maintenance issues if motion sensors fail to detect movement.

Additionally, the system provides real-time readings of LDR sensors to determine whether it is dark or light. The LED column in the system displays the status of the LEDs on each pole, indicating whether they are "ON" or "OFF", and records the exact time of movement detection for each sensor. This data is then used to accurately calculate power consumption of the LEDs in an efficient manner.



Conclusion

After conducting thorough research and gathering all necessary data, our objective was to design and develop a working prototype that could simulate real-life scenarios and bring our ideas to life. Through our research, we found that the Intelligent LED system was the most effective solution for energy savings and cost reduction, which aligned with our project objective. As a result, we successfully met our goal of creating a practical and efficient intelligent system.

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